RABTECH ACADEMY / DATA ANALYTICS & BUSINESS INTELLIGENCE

Swiggy Operations & Delivery Intelligence

Executive Decision Report & Capstone Presentation

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Data Analytics & Business Intelligence

Dataset period: 01 Jan 2023 – 31 Dec 2023 | 5,000 order records across 5 cities

Executive Summary

- 5,000 unique orders generated 3,363,056 revenue units and 438,302.45 profit units in the supplied dataset.
- Average order value: 672.61; average delivery time: 35.0 minutes; average customer rating: 4.00/5.
- 85.7% of records are marked Delivered; 8.0% are marked Cancelled.
- Mumbai leads city revenue; Bangalore has the highest cancellation rate and is the first city to investigate.

Decision: focus first on cancellation reduction in Bangalore, then test peak-period operational improvements.

Business Question & Objectives

- Can city, category, and time-slot patterns show where operational attention may have the highest value?
- How do order status, delivery duration, revenue, profit, and customer ratings vary across segments?
- Objective 1 — establish a consistent KPI baseline from the dataset.
- Objective 2 — identify high-priority segments without assuming correlation proves cause.
- Objective 3 — recommend practical pilots with measurable outcomes and guardrails.

Scope: this is an analysis of the supplied project dataset, not a statement about Swiggy’s live business performance.

Dataset Profile

- Rows: 5,000 | Unique Order IDs: 5,000 | Date range: 01 Jan 2023 – 31 Dec 2023.
- Dimensions: 5 cities, 6 food categories, 4 time slots, 4 order statuses.
- Fields: OrderID, OrderDate, City, Category, Revenue, Profit, DeliveryTimeMin, Status, PartnerID, CustomerRating, DistanceKM, TimeSlot.
- Data quality checks: 0 missing cells, 0 fully duplicated rows, 0 duplicate Order IDs.
- Dataset appears transaction-level; interpret revenue/profit as dataset units unless the source owner confirms currency and accounting definitions.

The report is based on the matching Swiggy_Dashboard_Dataset.xlsx found with the Power BI project.

Methodology & KPI Definitions

- Profile schema, missingness, duplicate rows, unique Order IDs, and date coverage.
- Aggregate order count, revenue, profit, average order value, average delivery duration, rating, and status rates.
- Compare city, category, time slot, and month; inspect variation rather than only totals.
- Revenue = sum of Revenue; Profit = sum of Profit; AOV = Revenue ÷ unique orders.
- Status rate = count of a status ÷ all order records; averages use the available field values.

The calculations below are reproducible from the supplied workbook; no live operational data was used.

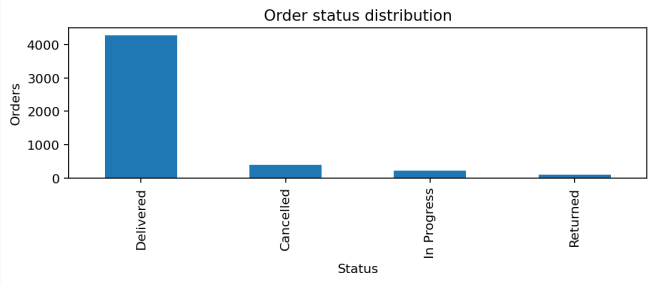
Executive KPI Scorecard

- Total orders: 5,000
- Total revenue: 3,363,056 dataset units
- Total profit: 438,302.45 dataset units | Profit-to-revenue ratio: 13.03%
- Average order value: 672.61 dataset units
- Average delivery time: 34.98 minutes | Median: 35 minutes
- Average customer rating: 4.00/5 | Average distance: 7.95 km

These values describe the provided 2023 project dataset and should be refreshed if the underlying data changes.

Order Status & Fulfilment

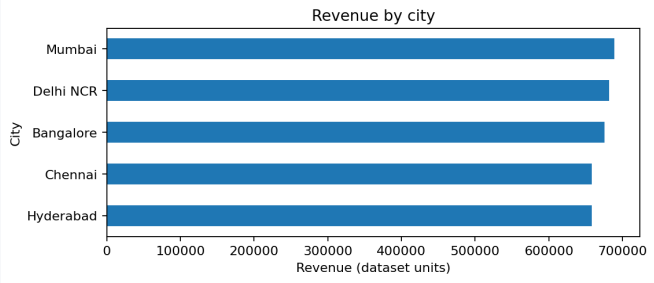
- Delivered: 4,285 (85.70%).
- Cancelled: 401 (8.02%).
- In Progress: 222 (4.44%).
- Returned: 92 (1.84%).
- Do not group In Progress or Returned with Delivered when reporting successful fulfilment.
- Recommended next step: review cancellation reasons and operational stage for every non-delivered order, if reason codes are available.



The current dataset has order status but no cancellation reason field; root causes require additional evidence.

City Performance

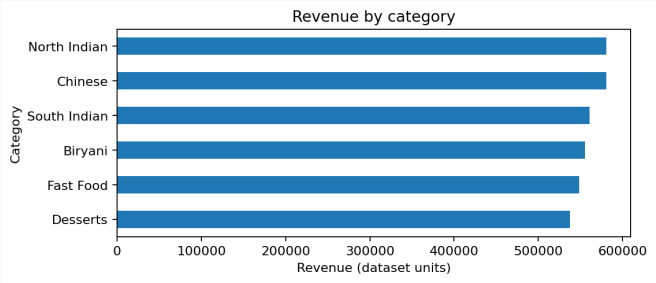
- Mumbai has the highest total revenue: 689,204 units from 1,032 orders.
- Bangalore has the highest cancellation rate: 10.11%.
- Delhi NCR has the highest Delivered share: 87.53%.
- City revenue range: 658,144 to 689,204 units; totals are relatively close across cities.
- Interpret totals alongside order counts and status rates so larger cities do not automatically appear more efficient.



Priority: investigate Bangalore cancellation patterns first, while comparing its order mix and time slots with a lower-cancellation city.

Revenue, Profit & Category Mix

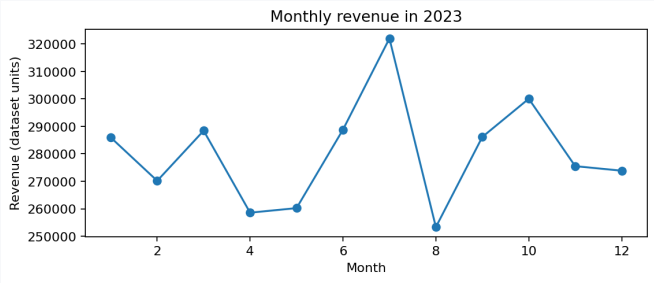
- Highest-revenue category: North Indian (580,480 revenue units).
- Highest-revenue category profit: 75,887.51 profit units in Chinese.
- Overall profit-to-revenue ratio: 13.03% based on the supplied Profit and Revenue fields.
- North Indian and Chinese categories are the top two by revenue and are close in total value.
- Before promotional decisions, validate whether Profit includes all discounts, refunds, delivery costs, and platform costs.



Treat profit-to-revenue as a dataset ratio, not audited corporate margin.

Time Slot & Monthly Trends

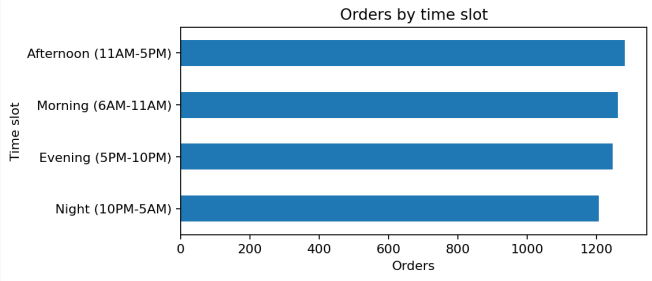
- Afternoon (11AM-5PM) has the highest order volume: 1,282 orders.
- Evening (5PM-10PM) has the longest average delivery time: 35.31 minutes.
- Highest-revenue month: July (321,958 units).
- Lowest-revenue month: August (253,353 units).
- Monthly variation should be interpreted as a pattern in this dataset; seasonality or cause cannot be established from one year alone.



Operational hypothesis: review evening delivery duration, while planning staffing around observed demand rather than assumptions.

Delivery & Customer Experience

- Average delivery duration is 34.98 minutes (median 35 minutes).
- Average rating is 4.00/5.
- Delivery time and rating have a very weak linear correlation in this dataset (r = -0.03); this does not support claiming that time alone explains ratings.
- Segment analysis should consider distance, city, time slot, status, and sample size together.
- Use lower-rating and longer-duration cases as investigation queues, not as proof of a single root cause.



Collect issue reasons, promised delivery time, actual delivery time, and customer feedback themes to improve diagnosis.

Risk Areas & Data Limitations

- Revenue/profit units and accounting definitions are not explicitly documented in the workbook.
- Dataset contains only one calendar year; long-term trend and seasonality claims are limited.
- Status labels alone do not explain cancellations, returns, or in-progress orders.
- Correlation is not causation; segment differences may reflect mix and volume.
- Do not present these sample-data results as current or official Swiggy company-wide metrics.

Mitigation: document definitions, add reason codes and service-level targets, and refresh the dashboard on a defined cadence.

Recommendation 1 — Reduce Cancellations

- Start with Bangalore, whose cancellation rate is 10.11% in the dataset.
- Break cancellations down by time slot, category, partner, and order stage; validate sample sizes.
- Pilot one targeted intervention at a time (for example, confirmation checks or availability accuracy) only after root cause is identified.
- Compare cancellation rate and completed orders against a comparable baseline segment.
- Owner: City Operations Lead | Review cadence: weekly.

Success measure: a sustained reduction in cancellation rate without reducing completed orders or customer rating.

Recommendations 2 & 3 — Capacity & Category Mix

- 2 — Peak capacity: review the evening time slot, which has the longest average delivery duration ({slot.loc[slow_slot,'Delivery']:.2f} minutes); test capacity/process changes in a controlled pilot.
- 3 — Revenue mix: compare category revenue and profit together; test targeted offers with a holdout group instead of broad discounting.
- Set a baseline and guardrails before each pilot: delivery duration, cancellations, ratings, revenue, and profit.
- Scale only when the improvement repeats across comparable periods and does not damage guardrail KPIs.

Owners: Operations Lead for capacity; Commercial/Category Lead for mix and promotion tests.

ROI Scenario & Measurement Framework

- Use a transparent pilot model: Net benefit = verified incremental benefit – implementation cost.
- ROI (%) = (verified incremental benefit – implementation cost) ÷ implementation cost × 100.
- Illustrative only: ■100,000 incremental benefit and ■40,000 cost gives ■60,000 net benefit and 150% ROI for the stated period.
- Those example amounts are not calculated from the dataset and must not be reported as actual Swiggy savings.
- Record one-off costs, recurring costs, measurement window, baseline, and control comparison.

No defensible ROI forecast is possible until intervention cost and incremental benefits are measured.

Implementation Plan & Final Decision

- Week 1 — validate KPI definitions and confirm data currency, status meanings, and cost definitions.
- Week 2 — diagnose cancellation patterns and select a city/time-slot pilot with a baseline/control.
- Weeks 3–4 — run cancellation and capacity pilots; record weekly operational and customer metrics.
- Week 5 — quantify verified benefit, cost, and guardrail movement.
- Week 6 — scale, revise, or stop; document results and refresh the Power BI dashboard.

Final recommendation: prioritise cancellation diagnosis in Bangalore, monitor evening delivery duration, and use category-level profitability for targeted decisions.